

Engineering Mathematics

Comprehensive Study Notes for GATE Computer Science

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Chapter 1

Matrices

1.1 Singular and Non-Singular Matrices

Definition 1.1.1. A square matrix A is said to be **singular** if $|A| = 0$, and **non-singular** if $|A| \neq 0$.

1.2 Properties of Matrix Addition and Multiplication

Let A , B , and C be matrices. The following properties hold:

- (i) $A + B = B + A$ (Commutative)
- (ii) $(A + B) + C = A + (B + C)$ (Associative)
- (iii) $AB \neq BA$ (Not Commutative)
- (iv) $(AB)C = A(BC)$ (Associative)
- (v) $A(B + C) = AB + AC$ (Distributive)

1.3 Transpose of a Matrix

Definition 1.3.1. If $A = [a_{ij}]_{m \times n}$, then $A' = [b_{ij}]_{n \times m}$ where $b_{ij} = a_{ji}$.

1.3.1 Properties of Transpose of a Matrix

- (i) $(A')' = A$
- (ii) $(A + B)' = A' + B'$
- (iii) $(AB)' = B'A'$

1.4 Symmetric Matrix

Definition 1.4.1. A matrix A is **symmetric** if $A' = A$.

1.5 Skew-Symmetric Matrix

Definition 1.5.1. A matrix A is **skew-symmetric** (or anti-symmetric) if $A' = -A$. If A and B are skew-symmetric matrices, then $A' = -A$ and $B' = -B$, so $(AB - BA)' = -AB + BA$ (skew-symmetric).

1.6 Orthogonal Matrix

Definition 1.6.1. A matrix A is **orthogonal** if $AA' = A'A = I$.

1.7 Idempotent Matrix

Definition 1.7.1. A matrix A is **idempotent** if $A^2 = A$.

1.8 Involutory Matrix

Definition 1.8.1. A matrix A is **involutory** if $A^2 = I$.

Theorem 1.8.1. Every square matrix can be uniquely expressed as the sum of a symmetric matrix and a skew-symmetric matrix:

$$A = \frac{1}{2}(A + A')(\text{symmetric}) + \frac{1}{2}(A - A')(\text{skew-symmetric})$$

1.9 Adjoint of a Square Matrix

Definition 1.9.1. If $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, then the adjoint of A is

$$\text{Adj } A = \text{Transpose of } \begin{bmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{bmatrix} = \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix}$$

where A_i, B_i, C_i are cofactors of a_i, b_i, c_i respectively.

1.9.1 Properties of Adjoint

$$(i) \quad A(\text{Adj } A) = (\text{Adj } A)A = |A|I_n$$

$$(ii) \quad \text{Adj}(AB) = (\text{Adj } B) \cdot (\text{Adj } A)$$

1.10 Inverse of a Square Matrix

Definition 1.10.1. The inverse of a matrix A is given by

$$A^{-1} = \frac{\text{Adj } A}{|A|}, \quad |A| \neq 0$$

1.10.1 Properties of Inverse

- (i) $(A^{-1})' = A$
- (ii) $(AB)^{-1} = B^{-1}A^{-1}$
- (iii) $(A')^{-1} = (A^{-1})'$
- (iv) Only a non-singular square matrix can have an inverse.

Example: Find A^{-1} if $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$.

Solution:

$$\text{Adj } A = \begin{bmatrix} 1 & -2 & -2 \\ -1 & 3 & 3 \\ 0 & -4 & -3 \end{bmatrix}' = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

$$|A| = 3(-3 + 4) + 3(2) - 4(-1) = 1$$

$$A^{-1} = \frac{\text{Adj } A}{|A|} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

1.11 Elementary Matrices

Definition 1.11.1. The matrix obtained from a unit matrix I by subjecting it to one of the elementary operations is called an **elementary matrix**.

1.12 Rank of a Matrix

Definition 1.12.1. Let A be any $m \times n$ matrix. The rank is defined such that:

- A has square sub-matrices of different orders. The determinants of these square sub-matrices are called **minors** of A .
- A matrix is said to be of **rank** r , if
 - (i) It has at least one non-zero minor of order r
 - (ii) All its minors of order $(r + 1)$ or higher than r are zero

The rank of $A = r$ is written as $\rho(A) = r$.

If A is a non-singular $n \times n$ matrix, then $\rho(A) = n$.

1.12.1 Echelon Form Method of Finding Rank

In the echelon form of a matrix, each of the first r elements of the leading diagonal is 1, and every element below this diagonal is 0. The rank of the matrix is equal to the number of non-zero diagonal elements when it has been reduced to echelon form.

Chapter 2

System of Linear Equations

2.1 Solution of a System of Linear Equations

A system of equations having no solution is called an **inconsistent system** of equations.

A system of equations having one or more solutions is called a **consistent system** of equations.

2.1.1 For a System of Non-Homogeneous Linear Equations $AX = B$

- (i) If $\rho[A : B] \neq \rho(A)$, the system is inconsistent.
- (ii) If $\rho[A : B] = \rho(A) = \text{number of unknowns}$, the system has a unique solution.
- (iii) If $\rho[A : B] = \rho(A) < \text{number of unknowns}$, the system has an infinite number of solutions.

2.1.2 For a System of Homogeneous Linear Equations $AX = 0$

- (i) $X = 0$ is always a solution (Trivial Solution).
- (ii) If $\rho(A) = \text{number of unknowns}$, the system has only the trivial solution.
- (iii) If $\rho(A) < \text{number of unknowns}$, the system has an infinite number of non-trivial solutions.

A homogeneous system is always consistent.

2.2 Linear Dependence and Linear Independence of Vectors

Definition 2.2.1. A set of n -vectors $X_1, X_2, \dots, X_r$ in V is said to be **linearly dependent** if there exist r scalars (numbers) $k_1, k_2, \dots, k_r$, not all zero, such that

$$k_1X_1 + k_2X_2 + \dots + k_rX_r = 0$$

It is called **linearly independent** if every relation of the type

$$k_1X_1 + k_2X_2 + \cdots + k_rX_r = 0$$

implies $k_1 = k_2 = \cdots = k_r = 0$.

Chapter 3

Eigenvalues and Eigenvectors

3.1 Characteristic Equation

If A is a square matrix of order n , we can form the matrix $(A - \lambda I)$, where λ is a scalar and I is an identity matrix of order n . The determinant of this matrix equated to zero,

$$|A - \lambda I| = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix} = 0$$

is called the **characteristic equation** of A .

The roots of this equation are called the **characteristic roots** or **eigenvalues** of A .

3.2 Eigenvectors and Eigenvalues

Definition 3.2.1. A matrix A of size $(n \times n)$ has a column vector X of size $(n \times 1)$ called the **Eigen Vector** of A , if

$$AX = \lambda X$$

$$AX - \lambda X = 0$$

$$(A - \lambda I)X = 0$$

where λ is a nonzero scalar.

The characteristic roots of equation $(A - \lambda I) = 0$ are called the Eigen Values.

Example: Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$.

Solution: The characteristic equation of the given matrix is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -2 - \lambda & 2 & -3 \\ 2 & 1 - \lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix} = 0$$

$$(\lambda + 3)(\lambda + 3)(\lambda - 5) = 0 \implies \lambda = -3, -3, 5$$

Corresponding to $\lambda = -3$, the eigenvectors are given by $(A + 3I)X_1 = 0$:

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = k_1, \quad x_2 = 2k_1, \quad x_3 = -k_1$$

$$X_1 = k_1 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

For $\lambda = 5$, let $X_2 = k_2$ and $x_1 = k_2$, then $x_2 = 2k_2$, $x_3 = -k_2$.

$$X_2 = k_2 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

3.3 Properties of Eigenvalues and Eigenvectors

- (1) The sum of the eigenvalues of a matrix A is equal to the trace of A .
- (2) The product of the eigenvalues of a matrix A is equal to its determinant.
- (3) If λ is an eigenvalue of an orthogonal matrix, then $1/\lambda$ is also its eigenvalue.
- (4) The eigenvalues of an idempotent matrix are either zero or unity.
- (5) The trace or spur of a square matrix is the sum of its diagonal elements.

Chapter 4

Limits, Continuity and Differentiability

4.1 Trigonometry

4.1.1 Basic Identities

$$(1) \sin^2 \theta + \cos^2 \theta = 1 \quad \text{All } (+), 180^\circ \sin(+), \text{Others } (-)$$

$$(2) 1 + \tan^2 \theta = \sec^2 \theta \quad \tan, \cot (+), \text{Others } (-)$$

$$(3) 1 + \cot^2 \theta = \csc^2 \theta \quad 270^\circ \cos(+), \text{Others } (-)$$

$$(4) \sec \theta = 1/\cos \theta, \quad \csc \theta = 1/\sin \theta$$

$$(5) \tan \theta = \sin \theta / \cos \theta, \quad \cot \theta = \cos \theta / \sin \theta$$

4.1.2 Sum and Difference Formulas

$$(1) \sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$(2) \sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$(3) \cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$(4) \cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$(5) \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}, \quad \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

4.1.3 Product-to-Sum Formulas

$$(1) 2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$(2) 2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

$$(3) 2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$(4) 2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

4.1.4 Sum-to-Product Formulas

- (1) $\sin C + \sin D = 2 \sin \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$
- (2) $\sin C - \sin D = 2 \cos \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$
- (3) $\cos C + \cos D = 2 \cos \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$
- (4) $\cos C - \cos D = -2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$

4.1.5 Double Angle Formulas

- (1) $\sin 2x = 2 \sin x \cos x$
- (2) $\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$
- (3) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

4.1.6 Hyperbolic Functions

- (1) $\sin 2x = \frac{2 \tan x}{1 + \tan^2 x}$
- (2) $\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$

4.2 Continuity and Differentiability

Definition 4.2.1. A function is **continuous** if its graph is a single unbroken curve with no holes or jumps.

Definition 4.2.2. A function is **differentiable** if its graph is relatively smooth and does not contain any breaks or bends.

Note: A differentiable function is always continuous, but a continuous function need not be differentiable.

4.3 Differentiation

4.3.1 Important Formulas

- (i) $\frac{d}{dx}(x^n) = nx^{n-1}$
- (ii) $\frac{d}{dx}(e^x) = e^x$
- (iii) $\frac{d}{dx}(a^x) = a^x \log_e a$
- (iv) $\frac{d}{dx}c = 0$ where c is a constant
- (v) $\frac{d}{dx} \cos x = -\sin x$
- (vi) $\frac{d}{dx}(\tan x) = \sec^2 x$
- (vii) $\frac{d}{dx} \sec x = \sec x \tan x$

$$(viii) \quad \frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$(ix) \quad \frac{d}{dx} \cot x = -\csc^2 x$$

$$(x) \quad \frac{d}{dx}(\log_e x) = \frac{1}{x}$$

$$(xi) \quad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$(xii) \quad \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$(xiii) \quad \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$(xiv) \quad \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$(xv) \quad \frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$(xvi) \quad \frac{d}{dx}(\csc^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}$$

$$(xvii) \quad \frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

$$(xviii) \quad \frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

$$(xix) \quad \frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$$

4.4 Limits

4.4.1 Important Expansions to be Used

$$(i) \quad (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$$

$$(ii) \quad \left(\frac{x^n - a^n}{x - a} \right) = (x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-1})$$

$$(iii) \quad e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$(iv) \quad a^x = 1 + x \log_e a + \frac{(x \log_e a)^2}{2!} + \dots$$

$$(v) \quad \log(1+x) = x - \frac{x^2}{2!} + \frac{x^3}{3!} - \frac{x^4}{4!} + \dots$$

$$(vi) \quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$(vii) \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$(viii) \quad \tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$$

4.4.2 Some Important Theorems on Limits

- (i) $\lim_{x \rightarrow a} \left(\frac{x^n - a^n}{x - a} \right) = na^{n-1}$ where $a > 0$
- (ii) $\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right) = 1, \quad \lim_{a \rightarrow 0} \left(\frac{a^x - 1}{x} \right) = \log_e a$
- (iii) $\lim_{x \rightarrow \infty} (1 + x)^n = e$
- (iv) $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$
- (v) $\lim_{x \rightarrow \infty} \left(\frac{a^x}{x} \right) = 0$
- (vi) $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) = 1, \quad \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right) = 1$

Chapter 5

Integration

5.1 Fundamental Integration Formulas

$$(i) \int x^n dx = \frac{x^{n+1}}{(n+1)} + C$$

$$(ii) \int \frac{1}{x} dx = \log x + C$$

$$(iii) \int e^x dx = e^x + C$$

$$(iv) \int a^x dx = \frac{a^x}{\log_e a} + C$$

$$(v) \int \sin x dx = -\cos x + C$$

$$(vi) \int \cos x dx = \sin x + C$$

$$(vii) \int \sec^2 x dx = \tan x + C$$

$$(viii) \int \csc^2 x dx = -\cot x + C$$

$$(ix) \int \sec x \tan x dx = \sec x + C$$

$$(x) \int \csc x \cot x dx = -\csc x + C$$

$$(xi) \int \cot x dx = \log(\sin x) + C$$

$$(xii) \int \tan x dx = -\log(\cos x) + C$$

$$(xiii) \int \sec x dx = \log(\sec x + \tan x) + C$$

$$(xiv) \int \csc x dx = \log(\csc x - \cot x) + C$$

$$(xv) \int \frac{dx}{a^2 - x^2} = \sin^{-1} \left(\frac{x}{a} \right) + C$$

$$(xvi) \int \frac{-1}{a^2 - x^2} dx = \cos^{-1} \left(\frac{x}{a} \right) + C$$

$$(xvii) \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$(xviii) \int \frac{-1}{a^2 + x^2} dx = \frac{1}{a} \cot^{-1} \left(\frac{x}{a} \right) + C$$

$$(xix) \int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \left(\frac{x}{a} \right) + C$$

$$(xx) \int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \csc^{-1} \left(\frac{x}{a} \right) + C$$

5.2 Integration by Substitution

$$\begin{aligned} I &= \int f(g(x)) \cdot g'(x) dx \\ \text{Let } g(x) &= t \\ \implies g'(x) dx &= dt \\ \implies I &= \int f(t) dt \end{aligned}$$

Note: If $\int f(x) dx = g(x) + c$, then $\int f(ax + b) dx = \frac{1}{a}g(ax + b) + c$.

5.3 Integration by Parts

$$\int uv dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$$

Chapter 6

Numerical Methods

6.1 Numerical Solution of Algebraic Equations

6.1.1 Bisection Method

Let the function $f(x)$ be continuous between a and b , and let $f(a)$ be negative and $f(b)$ be positive. Then the first approximation to the root is

$$x_1 = \frac{1}{2}(a + b)$$

If $f(x_1)$ is negative, $x_1 = \frac{1}{2}(x_1 + b)$ and so on.

If $f(x_1)$ is positive, $x_2 = \frac{1}{2}(a + x_1)$ and so on.

6.1.2 Secant Method

$$x_{n+1} = x_n - \left[\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \right] f(x_n)$$

6.1.3 Newton-Raphson Method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

6.1.4 Regula-Falsi Method (Always Converges)

$$x_{n+1} = x_0 - \left[\frac{x_n - x_0}{f(x_n) - f(x_0)} \right] f(x_n)$$

Convergence Rate:

Newton-Raphson > Secant > Regula-Falsi (Quadratic) > (Linear)

- Newton's method does not always converge.
- Order of convergence of secant method is 1.618
- Order of convergence of Newton's method is 2
- Order of convergence of Regula-Falsi method is 1
- If the initial values are not close enough to the root, then there is no guarantee that the secant method converges.

6.2 Numerical Integration

6.2.1 Trapezoidal Rule

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \cdots + y_{n-1})]$$

6.2.2 Simpson's One-Third Rule

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \cdots + y_{n-1}) + 2(y_2 + y_4 + \cdots + y_{n-2})]$$

6.2.3 Simpson's Three-Eighth Rule

$$\int_{x_0}^{x_n} f(x) dx = \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \cdots + y_{n-2}) + 2(y_3 + y_6 + \cdots + y_{n-3})]$$

Here h is the interval, and n is the number of intervals.

- $y_0 = f(x_0)$, $y_1 = f(x_0 + h)$, $y_2 = f(x_0 + 2h)$ and so on
- For Simpson's One-Third rule, n is even
- For Simpson's Three-Eighth rule, n should be a multiple of 3

6.2.4 Error in Trapezoidal Rule

$$\text{Error} \leq \frac{(b-a)^3}{12n^2} \max |f''(x)|$$

or

$$\text{Error} \leq \frac{(b-a)h^2}{12} \max |f''(x)|$$

6.2.5 Error in Simpson's Rule

$$\text{Error} \leq \frac{(b-a)h^4}{180} \max |f^{(4)}(x)|$$

Here, $a = x_0$, $b = x_0 + nh$, $h = \frac{b-a}{n}$, and $f'(x)$ and $f''(x)$ refer to the integrand taken on $[a, b]$.

- Simpson's rule provides exact results for any polynomial of degree three or less.
- Trapezoidal rule gives exact results for any polynomial of degree one.